

NAIROBI JOMO KENYATTA INTL, Kenya

WMO: 637400

Lat: 1.319S

Lon: 36.928E

Elev: 1624

StdP: 83.28

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	11.7	4.0	6.1	26.0	5.8	7.0	24.1	8.1	21.8	7.1	21.8	2.0	240	0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	11.7	29.2	16.0	28.5	16.0	27.8	16.0	19.1	23.7	18.7	23.3	18.3	23.0	6.0	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	15.8	19.8	17.4	15.2	19.4	17.1	15.0	19.2	61.9	24.2	60.5	23.5	59.3	22.6	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.3	8.3	7.5	7.8	31.4	1.0	1.5	7.1	32.5	6.6	33.4	6.0	34.2	5.3	35.3
			7.0	20.3	0.9	0.7	6.4	20.9	5.8	21.3	5.4	21.7	4.7	22.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.9	20.6	21.3	21.6	20.8	19.7	18.5	17.7	18.2	19.5	20.6	20.1	20.2	(6)
	DBStd	1.67	1.09	1.19	1.19	0.96	1.02	1.21	1.37	1.29	1.37	1.18	1.08	0.99	(7)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD18.3	70	1	0	0	0	2	12	29	19	5	1	1	1	(9)
	CDD10.0	3611	328	316	360	323	302	254	238	254	286	329	304	316	(10)
	CDD18.3	639	71	82	102	73	46	16	9	14	41	72	55	58	(11)
	CDH23.3	3663	518	668	705	276	114	63	73	106	300	427	172	241	(12)
	CDH26.7	459	62	134	146	18	2	1	2	3	28	48	6	9	(13)
(14) Wind	WSAvg	3.5	4.1	4.2	4.2	3.3	2.8	2.6	2.7	2.8	3.1	3.8	3.9	4.2	(14)
(15) Precipitation	PrecAvg	736	59	48	75	138	104	30	12	14	18	44	123	92	(15)
	PrecMax	1291	367	289	203	348	372	226	63	63	78	162	484	331	(16)
	PrecMin	327	0	0	1	1	3	0	0	0	0	0	26	6	(17)
	PrecStd	227	71	55	56	78	83	37	15	15	19	38	87	76	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.2	30.2	30.6	28.4	26.8	26.2	26.9	27.1	28.9	29.0	27.2	27.7	(19)
		MCWB	15.5	15.6	16.0	16.7	17.2	16.8	16.3	16.1	15.8	16.3	16.7	16.0	(20)
	2%	DB	28.1	29.2	29.2	27.0	25.3	25.0	25.1	25.8	27.2	28.0	26.2	26.8	(21)
		MCWB	15.5	15.7	16.3	17.3	17.3	16.5	15.8	15.7	15.7	15.9	16.7	16.3	(22)
	5%	DB	27.2	28.2	28.2	26.0	24.7	24.0	24.0	24.4	26.2	27.0	25.1	25.8	(23)
		MCWB	15.9	15.7	16.3	17.3	17.3	16.4	15.7	15.5	15.7	16.1	16.9	16.4	(24)
	10%	DB	26.2	27.2	27.1	25.0	23.8	22.9	22.7	23.1	25.2	25.9	24.2	24.9	(25)
		MCWB	16.1	16.0	16.7	17.4	17.2	16.1	15.3	15.3	15.6	16.0	17.0	16.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.9	19.2	19.2	19.6	19.2	18.5	17.5	17.6	18.0	18.4	18.9	19.1	(27)
		MCDB	24.0	24.6	24.7	24.4	23.2	23.2	23.7	23.9	25.0	24.3	22.9	23.4	(28)
	2%	WB	18.2	18.5	18.6	19.0	18.6	17.7	16.7	16.7	17.1	17.8	18.2	18.4	(29)
		MCDB	23.3	23.7	24.1	23.3	22.6	22.5	22.8	22.9	24.1	23.4	22.4	22.7	(30)
	5%	WB	17.8	17.9	18.2	18.6	18.2	17.1	16.2	16.2	16.6	17.3	17.9	18.0	(31)
		MCDB	22.8	23.3	23.6	22.8	22.3	21.7	22.0	22.2	23.4	22.5	22.1	22.3	(32)
	10%	WB	17.3	17.4	17.8	18.2	17.8	16.6	15.7	15.7	16.2	16.9	17.6	17.6	(33)
		MCDB	22.3	22.9	23.1	22.4	21.8	21.1	20.9	21.3	22.7	21.9	21.5	21.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	12.9	11.7	9.6	9.0	9.6	10.3	10.4	12.1	11.4	8.9	9.9	(35)
		MCDBR	13.8	14.5	13.5	10.9	10.3	11.5	13.1	13.3	14.0	13.1	10.3	11.5	(36)
	5% WB	MCWBR	4.2	4.3	3.7	3.1	3.6	4.6	5.2	5.0	4.5	3.5	2.8	3.4	(37)
		MCWBR	10.5	11.5	10.8	9.1	8.6	10.0	11.2	11.3	12.3	10.8	8.6	8.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.371	0.399	0.381	0.329	0.327	0.365	0.379	0.364	0.345	0.356	0.352	0.364	(40)
		taud	2.437	2.337	2.417	2.629	2.625	2.467	2.389	2.422	2.480	2.456	2.508	2.476	(41)
	Ebn at Noon		960	934	941	970	948	897	889	923	961	963	972	962	(42)
		Edh at Noon	120	134	123	96	93	107	117	117	114	117	111	114	(43)
(44) All-Sky Solar Radiation	RadAvg		6.47	6.76	6.55	5.80	5.30	4.68	4.61	4.91	5.94	6.07	5.63	5.99	(44)
	RadStd		0.42	0.34	0.43	0.30	0.20	0.33	0.43	0.36	0.34	0.30	0.30	0.36	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.47	+0.69	N/A	+0.57	+0.39	+0.16	N/A	N/A	N/A	N/A
Regional (6 neighbors)		+0.47	+0.69	N/A	+0.57	+0.39	+0.16	N/A	N/A	N/A	N/A

Nomenclature: See separate page